Week 2 Reference Sheet: Variables and Data Types

CSC-125-201 | Use this as a reference when completing Homework 2

Homework Topics

- Creating variables with different data types
- Variable naming rules
- Checking types with `type()`
- Type conversion: `int()`, `float()`, `str()`
- User input with `input()`
- Basic arithmetic operations
- Debugging common errors

Quick Reference

Variables

```
# Creating variables
hero_name = "Spider-Man"        # string
power_level = 8.5               # float
years_active = 10               # integer
from_outer_space = True         # boolean

# Check data type
print(type(hero_name))          # <class 'str'>
print(type(power_level))        # <class 'float'>
```

The Four Data Types

Type	Stores	Example
String (`str`)	Text in quotes	`"Hello"`, `'blue'`
Integer (`int`)	Whole numbers	`42`, `-5`, `0`
Float (`float`)	Decimal numbers	`3.14`, `19.99`
Boolean (`bool`)	True/False	`True`, `False`

Note: `"100"` (with quotes) is a string. `100` is an integer. `100.0` is a float.

Variable Naming Rules

```
# Valid names
my_variable = 5
user_age = 25
hero_name = "Batman"
is_valid = True

# Invalid names (will cause errors)
# 2nd_variable = 5        # cannot start with number
# my-variable = 5         # no dashes
# my variable = 5         # no spaces
```

Type Conversion

```
# String to number
age_text = "25"
age_number = int(age_text)      # 25 (integer)
price_text = "19.99"
```

```
price_number = float(price_text) # 19.99 (float)

# Number to string
score = 100
score_text = str(score)         # "100" (string)

# Important: not all conversions work
# int("Hello") will cause a ValueError
```

User Input

```
# Getting text input
name = input("What's your name? ")

# Getting number input (must convert)
age = int(input("Enter your age: "))
height = float(input("Enter height: "))

# Using the input
print("Hello", name)
print("Next year you'll be", age + 1)
```

Basic Arithmetic

```
x = 10
y = 3

addition = x + y        # 13
subtraction = x - y     # 7
multiplication = x * y  # 30
division = x / y        # 3.333...
```

Common Errors

Error	Cause	Fix
SyntaxError	Missing quotes, parentheses	Check syntax carefully
NameError	Typo or undefined variable	Check variable spelling
TypeError	Adding string + number	Convert types first
ValueError	Invalid conversion	Ensure string is a valid number

Additional Resources

Variables and Data Types

LearnPython.org - Variables and Types
https://www.learnpython.org/en/Variables_and_Types
- Interactive tutorial with in-browser exercises

W3Schools - Python Variables
https://www.w3schools.com/python/python_variables.asp
- Variable creation, naming rules, and `type()`

Type Conversion

W3Schools - Python Casting
https://www.w3schools.com/python/python_casting.asp

- Using `int()`, `float()`, `str()`

Programiz - Type Conversion

https://www.programiz.com/python-programming/type-conversion-and-casting

- Explains implicit vs explicit type conversion

User Input

W3Schools - Python User Input

https://www.w3schools.com/python/python_user_input.asp

- `input()` always returns a string
- Combining `input()` with type conversion

Naming Conventions

Real Python - PEP 8 Guide

https://realpython.com/python-pep8/

- Python's official style guide for naming

Debugging

15 Common Python Errors - Better Stack

https://betterstack.com/community/guides/scaling-python/python-errors/

- Focus on the first 8 error types for this week